Advanced Calculus I Non-hand-in Assignment 5

This homework comes from textbook and will **NOT** be collected (這份作業完成後不需要繳交、評分。). But for practice,or quizzes or exams, please make sure that you have idea to solve the following questions by yourself. (這些題目是很重要的練習，小考、期中、期末考會有類似的題目。練習這些題目的過程中所逐漸形成我們對於所學到的知識的"看法"，會在考試中很有幫助。所以，請一定要了解如何解題與解題背後的想法是什麼。)

Please prove the following propositions are true. (請證明下列命題爲眞。)

Section 2.2.
2.9 Theorem (ii) (on page 60)
2.11 Theorem (on page 61)

Section 9.5.
Suppose that $\{x_k\}$ and $\{y_k\}$ are sequences of points in $\mathbb{R}^n$. Also, $\{x_k\}$ and $\{y_k\}$ converge. Please show that

$$\lim_{k\to\infty}(x_k + y_k) = \lim_{k\to\infty} x_k + \lim_{k\to\infty} y_k$$

第 5 週｜自主練習・課本題目補充

William R. Wade | An Introduction to Analysis | Fourth Edition, Global Edition

依教師 Non-hand-in Assignment 指定順序整理；以下為題目／命題敘述，不含解答。

Theorem 2.9 (ii)

Wade p.60

2.9 Theorem. [SQUEEZE THEOREM].
Suppose that $\{x_n\}$, $\{y_n\}$, *and* $\{w_n\}$ *are real sequences.*

i) *If* $x_n \to a$ *and* $y_n \to a$ *(the SAME a) as* $n \to \infty$, *and if there is an* $N_0 \in \mathbf{N}$ *such that*

$$x_n \leq w_n \leq y_n \quad \text{for } n \geq N_0,$$

then $w_n \to a$ *as* $n \to \infty$.

ii) *If* $x_n \to 0$ *as* $n \to \infty$ *and* $\{y_n\}$ *is bounded, then* $x_n y_n \to 0$ *as* $n \to \infty$.

本題僅指定 (ii)；此處保留定理的假設及完整敘述。

Theorem 2.11

Wade p.61

2.11 Theorem. *Let* $E \subset \mathbf{R}$. *If* E *has a finite supremum (respectively, a finite infimum), then there is a sequence* $x_n \in E$ *such that* $x_n \to \sup E$ *(respectively, a sequence* $y_n \in E$ *such that* $y_n \to \inf E$*) as* $n \to \infty$.

向量數列的極限加法

題目完整敘述已列於本檔第一頁原始作業，請證明該極限等式。